When Is Wealth Liquid? Gold and Household Financial Constraints in India*

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Abstract

Households can own substantial wealth and still lack the liquidity to sustain spending when income falls. We construct a joint household income and balance-sheet dataset for India by combining the Consumer Pyramids Household Survey with the All-India Debt and Investment Survey. We harmonize their common information and predict the missing assets and debts in the income and consumption panel. We then measure financial constraints and compare consumption responses to income shocks. Under the benchmark definition, 6.20 percent of CPHS households are poor hand-to-mouth and 40.74 percent are wealthy hand-to-mouth. Gold changes this classification. Counting it as illiquid wealth shifts households from the poor to the wealthy group; allowing for its potential collateral value further distinguishes households within that group. We apply the Blundell–Pistaferri–Preston method to annual income and consumption. Transitory-shock pass-through is 0.1874 for wealthy hand-to-mouth households with sufficient gold collateral capacity, compared with 0.2192 for constrained households without it. Permanent shocks produce a different ordering. The joint dataset thus connects the composition of wealth to financial constraints and consumption adjustment. It also illustrates how other countries can combine existing surveys to study household finance when no single source records all the required information.

Keywords: hand-to-mouth households; financial constraints; survey harmonization; household wealth; gold; income shocks; consumption insurance.

JEL codes: D12, D14, E21.

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1 Introduction

The wealth a household owns need not be the wealth it can use when income falls. A family may own its home, agricultural land, or jewellery while keeping little in a bank account. These assets add to net worth, but paying for food or meeting an unexpected bill may require a sale or a loan. How easily the household can raise cash determines whether it can sustain spending. We study this distinction in India, where gold helps reveal the boundary between wealth and liquidity.

We make three contributions to the study of household finance in India. First, we construct joint household income and balance sheets by combining two complementary surveys. This creates a resource for studying assets and debts alongside income and consumption. Second, we use those balance sheets to measure the shares of hand-to-mouth, or financially constrained, households. Third, we apply the method of Blundell et al. [2008], hereafter BPP, to compare annual consumption responses to income shocks across these groups. Gold plays a central role in both the classification and the comparison of responses.

The joint dataset brings together information that neither survey provides on its own. The Consumer Pyramids Household Survey (CPHS) follows income and expenditure over time but lacks the asset and debt amounts needed for the balance sheet. The All-India Debt and Investment Survey (AIDIS) records detailed wealth but not income. We harmonize their common variables and use AIDIS to predict liquid assets, illiquid assets, debts, and gold in CPHS. The resulting classification covers 129,906 CPHS households in 2019. Their observed income and consumption histories then allow us to study responses to income shocks. A complementary exercise predicts income in AIDIS, where wealth is observed.

Measuring financial constraints requires looking beyond poverty and net worth. Following Kaplan et al. [2014], we classify households using liquid wealth relative to income. Poor hand-to-mouth households have nonpositive net illiquid wealth; wealthy hand-to-mouth households have positive net illiquid wealth but little liquidity. We use financially constrained and hand-to-mouth interchangeably for households that meet this rule. The distinction builds on the link between consumption and liquidity emphasized by Zeldes [1989]. Because classification turns on small balances and zero holdings, our prediction method models ownership separately from amounts and evaluates both in AIDIS testing data.

Gold can support current spending even when a household wants to keep it. Jewellery combines family and cultural uses with saving, and it can remain attractive where formal financial products are difficult or costly to use [Singh et al., 2026, Household Finance Committee, 2017]. A sale

gives up the asset; a loan against it raises cash while creating an obligation to repay. We therefore examine gold as illiquid wealth, as liquid wealth, and as an illiquid asset with potential collateral value. The last case asks whether part of its value could bridge the household's measured liquid shortfall. Section 4.5 develops this context.

Most financially constrained households in our CPHS classification own positive illiquid wealth. With gold outside both wealth aggregates, 6.20 percent are poor hand-to-mouth and 40.74 percent are wealthy hand-to-mouth. Counting gold as illiquid lowers the poor share to 1.30 percent and raises the wealthy share to 45.64 percent. The non-hand-to-mouth share remains 53.06 percent because liquid wealth has not changed. Counting gold as fully liquid instead raises that share to 84.41 percent. Gold therefore matters both for the poor–wealthy distinction and for the measured prevalence of financial constraints.

Potential borrowing against gold provides an intermediate measure of usable wealth. We keep gold illiquid and ask whether net liquid wealth plus 75 percent of its value exceeds half of monthly income. Under this rule, 28.18 percent of all CPHS households are wealthy hand-to-mouth with sufficient gold collateral capacity. Another 17.46 percent are wealthy hand-to-mouth without it. Both groups have little financial liquidity, but their portfolios differ in the amount of cash that gold could potentially provide.

Households with different collateral capacity have different estimated responses to transitory income shocks. We use BPP to separate permanent and transitory income innovations and estimate their pass-through to consumption. Transitory pass-through is 0.1874 for collateral-capable wealthy hand-to-mouth households, 0.2192 for constrained households without sufficient capacity, and 0.2063 for non-hand-to-mouth households. Within the group without capacity, it is 0.2627 for households without positive non-gold illiquid wealth and 0.2085 for those with it. These are proportional consumption responses in the annual log specification. The estimates show why low liquid wealth alone does not describe how much spending adjusts to a temporary income change.

The ordering of responses also depends on the persistence of the shock. Collateral-capable wealthy households have the highest permanent-shock pass-through in the main comparison, reversing their position in the transitory comparison. In four alternative annual classifications, poor hand-to-mouth households consistently have the highest transitory-response point estimate. The ordering of wealthy and non-hand-to-mouth households changes when gold is counted as liquid. The findings therefore connect consumption adjustment to both the composition of household wealth and the type of income shock.

The joint dataset is a contribution beyond the particular household groups studied here. Household finance in emerging economies depends on the composition of balance sheets [Badarinza et al., 2019]. Bringing income, consumption, assets, and debts into one household-level resource makes those connections open to study. The Indian application also offers a practical route for other countries. Researchers and statistical agencies can identify complementary surveys, harmonize their common information, and adapt wealth definitions to local assets. They need not wait for one survey to collect every variable before beginning to measure financial constraints.

The paper proceeds from constructing the joint dataset to classifying households and comparing their responses. Section 2 describes the surveys, and Section 3 explains their harmonization. Section 4 presents wealth prediction, the classification rule, and the role of gold. Section 5 reports the BPP income-shock comparisons and robustness results. The appendices provide additional prediction diagnostics, classification checks, and annual estimates.

2 Data

Identifying financially constrained households requires distinguishing limited liquidity from low total wealth. Following Kaplan et al. [2014], we identify households with small liquid buffers or little remaining borrowing capacity relative to their income. We then distinguish poor hand-to-mouth households, whose net illiquid wealth is nonpositive, from wealthy hand-to-mouth households, whose net illiquid wealth is positive. Net wealth in each category subtracts the corresponding liabilities from assets. A household can therefore own a home or other illiquid assets and still be financially constrained. The poor–wealthy distinction describes its balance sheet, rather than its income level.

Implementing this classification requires income and detailed wealth information for each household. Income sets the scale of the liquidity threshold, while liquid assets and debts determine the household's position relative to it. Illiquid assets and their associated debts determine the poor–wealthy distinction. The Consumer Pyramids Household Survey (CPHS) supplies the income and consumption panel for our main analysis, but its asset indicators do not give the rupee amounts needed to construct net wealth. We therefore draw on the All-India Debt and Investment Survey (AIDIS) for those balance-sheet amounts. Section 3 explains how the surveys are harmonized, and Section 4.2 describes wealth prediction in CPHS.

2.1 Income and consumption in CPHS

CPHS is our main dataset because it follows household income and consumption over time. We use the income, expenditure, household-member, and asset-indicator modules released by the Centre for Monitoring Indian Economy [Centre for Monitoring Indian Economy, 2025]. Monthly records are assigned to their reference months and continuing household spells. The annual construction uses levels from 2014–2025 to form growth observations for 2015–2025. Household amounts are totals rather than per-capita values.

The classification cross-section pairs 2019 resources with ownership information near the AIDIS asset date. CPHS ownership indicators come from May–August 2018. The cross-section requires at least eight observed months in 2019 for income, expenditure, and household composition; continuous variables summarize observed months, while binary variables retain a positive report. The descriptive sample contains 129,908 households, and 129,906 enter the classification. The annual exercise attaches the resulting 2019 labels to each household's usable history.

CPHS income exceeds harmonized consumption at both the mean and the median. Table 1 reports mean monthly income of Rs. 20,884.94 and median income of Rs. 15,816.67. Mean monthly harmonized consumption is Rs. 12,074.48 and its median is Rs. 10,853.25. We use each household's observed monthly income to scale its liquidity threshold, rather than applying a common rupee cutoff.

2.2 Balance sheets in AIDIS

AIDIS supplies the asset and debt amounts that CPHS does not record. We use Visit 1 of the 2019 survey conducted by the National Statistical Office [National Statistical Office, 2021]. Asset records cover financial claims and physical assets, and loan records supply amounts, sources, and purposes. The classification sample contains 116,461 households. The observed asset and liability amounts enter the balance-sheet definitions before any model is fitted.

Reference dates distinguish wealth stocks from resource flows. Most AIDIS assets refer to 30 June 2018; cash in hand and the debt-purpose balances retain their survey-date fields. Asset and debt values are expressed in rupees per household. Consumption is expressed in rupees per household per month, combining purchased goods, self-production, in-kind receipts, and a monthly equivalent of annual durable expenditure.

The AIDIS descriptive sample records the scale of household consumption alongside household composition. Table 1 reports a mean household size of 4.30 and mean monthly consumption of Rs. 10,047.22. Median consumption is Rs. 8,250.00. The descriptive sample contains 116,456 households, a denominator distinct from the classification sample. These descriptive values characterize the source survey for wealth prediction.

Table 1: Harmonized household characteristics and resources

	CPHS mean	CPHS median	AIDIS mean	AIDIS median
Household size	3.98	4.00	4.30	4.00
Monthly harmonized consumption (Rs.)	12,074.48	10,853.25	10,047.22	8,250.00
Monthly total income (Rs.)	20,884.94	15,816.67	—	—
Households	129,908		116,456	

Notes: Survey-weighted descriptive statistics from the harmonized cross-sections. Resource flows are monthly rupees per household. Household size is a count. Income is not observed in AIDIS. The counts refer to the descriptive cross-sections; the classification and annual estimation samples are defined separately.

The cross-section measures how common financial constraints are, while the annual panel measures consumption responses. Survey-weighted summaries describe the harmonized cross-section and household-group shares. The annual estimates use an unweighted panel and report their own household and household-year counts. These counts need not reproduce the weighted prevalence shares. To build the joint dataset, we first make the common survey variables comparable.

3 Harmonization and balance-sheet measurement

We combine the surveys through common characteristics rather than matching individual households. AIDIS and CPHS have separate samples and no shared household identifier. We harmonize variables observed in both and use them to predict missing information: balance-sheet positions in CPHS and income in the complementary AIDIS exercise. The resulting joint dataset contains observed and predicted components, with each survey retaining its own households.

Prediction across surveys requires the common variables to measure the same economic object. A model fitted in one survey treats a predictor as the same variable when applied to the other. Differences in definition, reference period, or units can therefore become differences in predicted wealth. We align these features before fitting the models. The following subsections describe the common covariates, resource measures, asset indicators, and balance-sheet targets.

3.1 Reference periods and common covariates

We align stocks with stocks and flows with flows. The CPHS asset indicators are taken from the May–August 2018 wave, which brackets the AIDIS reference date for most asset stocks, 30 June 2018. CPHS income, consumption expenditure, and demographics summarize the observed 2019 reference months, while their AIDIS counterparts refer to the 2019 interview or the survey's stated recall window. CPHS binary indicators equal one if the household reports the item in any observed month of the relevant window. This provides a counterpart to the AIDIS positive-value indicator while retaining the distinction between reported ownership and a recorded rupee amount.

Common geographic categories make local information usable across the surveys. AIDIS records National Sample Survey district codes, while CPHS records CMIE district names. We use state and district information to construct a concordance. This prevents coding differences from being treated as differences in location.

Household size and age composition are computed on a common basis from the member roster. We use the number of members and the number in each of four age bands: under 18, 18–24, 25–64, and 65 or older. In CPHS these counts are averaged over the 2019 window. Religion, caste, education, and household employment type are mapped to common categories. Employment type distinguishes self-employment, regular wage or salaried work, casual labor, and other households, with agricultural and nonagricultural categories where the common coding supports them. For members aged 25–64, we record the number of men, the median age, and the highest education level. The latter two measures equal zero when the household has no member in this age band. These working-age characteristics describe the household's earning capacity alongside its size and age composition.

Table 2 reports weighted summary statistics for the common covariates. The two samples are similar in broad composition: about one-third of households are urban in each survey, and the median household has four members in both. AIDIS households are larger at the mean, with 4.30 members compared with 3.98 in CPHS. AIDIS households also contain more children: 1.36 members under 18 on average, compared with 1.05 in CPHS. The numbers of members aged 18–24 and 25–64 are close across the two surveys.

CPHS households have older and more educated working-age members on the measures used here. Across households, the median of the within-household median age is 42.83 years in CPHS and 39.50 years in AIDIS. The median highest education category among members aged 25–64 is secondary or higher secondary in CPHS and upper primary or middle school in AIDIS.

These characteristics enter the common predictor set alongside geography, household composition, religion, caste, and employment type.

Table 2: Common covariates: geography and demographics in CPHS and AIDIS

	CPHS				AIDIS			
Variable	N	Mean	Median	Zeros (%)	N	Mean	Median	Zeros (%)
Panel A. Geography								
Urban (1/0)	129,908	0.33	0.00	66.89	116,456	0.34	0.00	66.30
Panel B. Household size and age composition								
Household size	129,908	3.98	4.00	0.00	116,456	4.30	4.00	0.00
Members under 18	129,908	1.05	1.00	41.49	116,456	1.36	1.00	36.29
Members aged 18–24	129,908	0.56	0.00	56.71	116,456	0.58	0.00	61.19
Members aged 25–64	129,908	2.19	2.00	1.55	116,456	2.09	2.00	7.11
Members aged 65 or older	129,908	0.19	0.00	82.31	116,456	0.27	0.00	78.30
Panel C. Working-age members (25–64)								
Men aged 25–64	129,908	1.09	1.00	8.70	116,456	1.05	1.00	16.83
Median age (years)	129,908	42.62	42.83	1.55	116,456	38.04	39.50	7.11
Highest education[a]	129,908	4.47	5.00	1.55	116,456	3.64	4.00	7.11
Panel D. Religion, caste and employment type[a]								
Religion	129,905	1.22	1.00	0.00	116,456	1.29	1.00	0.00
Caste	129,905	4.43	3.00	0.00	116,456	4.24	3.00	0.00
Employment type	129,908	5.13	5.00	0.00	116,456	4.92	4.00	0.00

Notes: Household-level statistics for 2019. Each survey uses its own sampling weight: CMIE household weights for CPHS and NSS household multipliers for AIDIS. N is the unweighted number of households with a non-missing value; AIDIS households with a zero sampling weight are excluded. Means, medians, and shares of exact zeros are weighted. CPHS values are household averages over the observed 2019 reference months, so member counts need not be integers; AIDIS values refer to the 2019 interview. The median age and highest education in Panel C describe members aged 25–64 and equal zero when no member is in that band. [a]Categorical codes are as follows. Education: 1 = not literate, 2 = below primary, 3 = primary, 4 = upper primary/middle, 5 = secondary/higher secondary (including diploma), 6 = graduate (including diploma), 7 = post-graduate and above. Religion: 1 = Hindu, 2 = Muslim, 3 = Christian, 4 = Sikh, 5 = Jain, 6 = Buddhist, 7 = other. Caste: 1 = Scheduled Tribe, 2 = Scheduled Caste, 3 = Other Backward Classes, 9 = other. Employment type: 1 = urban self-employed, 2 = urban regular wage/salary, 3 = urban casual labor, 4 = rural self-employed in agriculture, 5 = rural self-employed in nonagriculture, 6 = rural regular wage/salary, 7 = rural casual labor in agriculture, 8 = rural casual labor in nonagriculture, 9 = retired/aged, 10 = other/not in the labor force.

3.2 Consumption and income

Our baseline consumption measure is intended to capture the resources a household consumes in a month rather than its cash outlays. In AIDIS it is the sum of purchased consumption, self-produced consumption, in-kind receipts, and one-twelfth of annual durable expenditure. In CPHS it equals total expenditure minus equated monthly installments and the health and general insurance premiums, plus income from self-production. Both measures are expressed in monthly rupees per household.

We subtract installments and insurance premiums because the released CPHS expenditure total includes them, whereas the concept of consumer expenditure that AIDIS follows does not. An installment repays a loan, and the good it financed is counted when it is purchased. Insurance premiums are recorded by AIDIS in a separate block of the questionnaire, outside consumer expenditure. CPHS reports separate health and general insurance premium items, which we remove from total expenditure. We add self-production because AIDIS values home-grown consumption directly, whereas CPHS records the same resource on the income side.

The consumption measures cover the same resources but are collected differently. AIDIS asks about usual monthly expenditure at one visit, whereas CPHS records itemized monthly expenditures, which we average over each household's observed 2019 months. Table 3 compares their distributions. Mean and median monthly consumption are higher in CPHS than in AIDIS. The table also reports the range of household consumption and the share with zero consumption in each survey.

Table 3: Monthly consumption in CPHS and AIDIS

Survey	Households	Minimum	Maximum	Mean	Median	Zeros (%)
CPHS	129,908	1,201.25	271,735.00	12,074.48	10,853.25	0.00
AIDIS	116,456	300.00	710,000.00	10,047.22	8,250.00	0.00

Notes: Household consumption in monthly current rupees. CPHS values are household means over the observed 2019 reference months; AIDIS values are usual monthly consumption at the 2019 interview. Means, medians, and shares of exact zeros use each survey's sampling weights (CMIE household weights for CPHS and NSS household multipliers for AIDIS). Household counts and the minimum and maximum are unweighted. Consumption includes purchased and self-produced resources and the monthly equivalent of durable expenditure, as defined in Section 3.2.

Income is observed only in CPHS. Total income includes every source recorded by CMIE, and Table 4 summarizes its distribution. We adopt total income as the baseline because it pairs a complete resource measure with the complete consumption measure. Observed CPHS income supplies the scale for identifying financially constrained households under the main hand-to-mouth rule. For the complementary AIDIS classification, the income-prediction model uses the common variables to supply the income absent from that survey.

Table 4: Monthly total income in CPHS

Measure	Households	Minimum	Maximum	Mean	Median	Zeros (%)
Total income	129,908	0.00	927,067.06	20,884.94	15,816.67	< 0.01

Notes: Total income includes every source recorded by CMIE and is expressed in monthly current rupees per household. Values are household means over the observed 2019 reference months. The mean, median, and share of exact zeros use CMIE household weights; the household count and the minimum and maximum are unweighted. A positive zero-income share below 0.01 percent is displayed as < 0.01 rather than rounded to zero.

Harmonized consumption helps predict wealth, while the annual spending measure captures consumption adjustment over time. The first is a common predictor that connects the sur-

veys. The second is twelve times median observed monthly total expenditure, as defined in Section 5.1. Keeping their roles distinct avoids treating the harmonized cross-section as the annual response sample.

3.3 Common financial and asset indicators

Both surveys record financial access, saving, borrowing, and asset ownership. These indicators describe portfolios and can help predict missing wealth. AIDIS infers holdings from positive asset values; CPHS asks about ownership and saving. We use an indicator only when the two questions describe the same attribute over comparable reference periods.

The surveys agree closely on some portfolio indicators and differ sharply on others. Table 5 reports weighted shares and identifies the indicators used in prediction. Livestock and car ownership are similar across surveys, and bank-account ownership is widespread in both. The table also compares borrowing by lender and purpose.

An asset stock and a saving decision are different economic objects. A household may own an asset without adding to it. A model fitted to positive asset holdings should therefore not treat an intention to save as the same variable. We compare the meaning of each survey question before using its indicator in prediction.

Fixed and recurring deposits illustrate the large differences in some saving indicators. They are reported by 61.45 percent of CPHS households and 2.62 percent of AIDIS households. The ordering is reversed for credit or debit cards, reported by 5.26 percent in CPHS and 59.00 percent in AIDIS. A model trained on one survey would treat each of these indicators as the same variable when applied to the other, despite the differences in how the instruments are recorded.

We exclude indicators whose definitions are not sufficiently comparable across surveys. These include dwelling and two-wheeler ownership and selected saving and insurance indicators, as shown in Table 5. Bank-account, car, and livestock ownership and borrowing from banks or moneylenders remain among the predictors. Excluding an indicator does not remove its asset from wealth: housing remains in illiquid wealth, and gold is predicted separately.

Table 5: Common financial and asset indicators in CPHS and AIDIS

	CPHS		AIDIS		In prediction models
Indicator	N	Share (%)	N	Share (%)	
Panel A. Financial access and saving instruments					
Bank account	129,908	99.54	116,456	97.35	Yes
Credit or debit card	129,908	5.26	116,101	59.00	No
Fixed or recurring deposits	129,908	61.45	116,456	2.62	No
Post-office savings	129,908	19.37	116,456	7.53	No
Provident fund	129,908	12.56	116,456	4.24	No
Life insurance, any policy	129,908	49.27	116,456	17.05	No
Life insurance held as savings	129,908	49.27	116,456	16.76	No
Savings in gold	129,908	98.91	116,456	82.48	No
Savings in a business	129,908	3.36	116,456	0.10	No
Listed shares	129,908	0.27	116,456	0.15	Yes
Mutual funds	129,908	0.85	116,456	0.34	Yes
Panel B. Borrowing					
Any outstanding borrowing	129,908	45.86	116,456	40.37	Yes
From banks	129,908	11.17	116,456	16.79	Yes
From self-help groups	129,908	5.88	116,456	7.61	Yes
From an NBFC or MFI	129,908	2.59	116,456	2.78	Yes
From moneylenders	129,908	15.77	116,456	10.29	Yes
From relatives or friends	129,908	11.31	116,456	7.55	Yes
For housing	129,908	5.05	116,456	6.53	Yes
For business or investment	129,908	4.57	116,456	16.00	No
For medical expenses	129,908	1.76	116,456	4.18	Yes
For education	129,908	1.67	116,456	1.55	Yes
For household expenditure	129,908	22.24	116,456	15.95	Yes
For repayment of debt	129,908	4.39	116,456	1.38	Yes
Panel C. Asset ownership					
Dwelling	129,908	99.47	116,456	82.90	No
Two-wheeler	129,908	63.24	116,456	40.09	No
Car	129,908	6.21	116,456	5.97	Yes
Tractor	129,908	2.91	116,456	2.08	Yes
Livestock	129,908	34.14	116,456	34.89	Yes
Computer	129,908	6.97	116,456	0.70	No

Notes: Household-level shares for which each indicator equals one, in percent, using each survey's sampling weight (CMIE household weights for CPHS and NSS household multipliers for AIDIS). N is the unweighted number of households with a non-missing value. AIDIS saving and asset indicators identify a positive value in the corresponding asset schedule, with financial assets dated 30 June 2018; borrowing indicators refer to loans outstanding at the 2019 interview. CPHS indicators equal one if the household reports the item in any observed month of the matching reference period described in Section 3.1. The last column shows whether the indicator enters the common predictor set for income and balance sheets. Excluding an ownership indicator does not remove that asset from the wealth aggregates. NBFC denotes a nonbank financial company and MFI a microfinance institution.

3.4 Balance-sheet components

The balance sheet separates financial liquidity from other wealth before subtracting debt. Let A^L and D^L denote liquid assets and liquid liabilities, and let A^I and D^I denote illiquid assets and their associated liabilities. Net positions are

$$m = A^L - D^L, \qquad a = A^I - D^I. \tag{1}$$

Cash, deposits, and the financial claims assigned to the liquid category enter A^L. Land, buildings, and the longer-term claims assigned to the illiquid category enter A^I. Housing liabilities are netted against illiquid wealth; the other debt assignments follow the corresponding liquid category.

We predict gold separately so that we can distinguish its wealth and liquidity roles. Let g denote its predicted value. The benchmark leaves gold outside m and a; the gold-illiquid definition uses $a+g$, and the gold-liquid definition uses $m+g$. We exclude life-insurance sum assured because it is a contingent benefit. The separate dwelling prediction is used for validation and is not added again to aggregate illiquid wealth. These definitions give us the components needed to predict household portfolios and classify financial constraints.

4 Classifying financially constrained households

We use the joint balance sheets to identify households with little liquidity between income receipts. A financially constrained, or hand-to-mouth, household can have low liquid wealth without having low income or low total wealth. We state the classification rule, explain wealth prediction in CPHS, and report the resulting household shares. We then examine how gold changes the classification and separates households by potential collateral capacity.

4.1 The classification rule: liquid and illiquid wealth

The two-asset framework distinguishes resources that can finance current spending from wealth that is costly to access. In Kaplan et al. [2014], a liquid asset can be drawn on at any time but earns a low return, whereas an illiquid asset earns a higher return but can be accessed only at a transaction cost. When the return differential is large enough, a household places its savings in the illiquid asset and holds no liquid balances. Such a household spends its income each

period despite owning substantial wealth. The wealthy hand-to-mouth category captures this distinction between wealth ownership and available liquidity.

Taking this definition to survey data requires an inference about balances between income receipts. The theory refers to liquid balances at the end of the pay period, whereas surveys report average balances or balances at the interview date. Kaplan et al. [2014] assume that income is paid into the liquid account at the start of the pay period and consumed at a constant rate. A household that exhausts its income by the end of the period holds half that income on average. We apply this logic to monthly income, so a positive observed liquid balance can still be consistent with living from one income receipt to the next.

The implemented rule covers both small nonnegative liquid balances and borrowers near their credit limit. Let $y > 0$ be monthly household income, m net liquid wealth, a net illiquid wealth, and $\bar{m}$ an assumed unsecured borrowing limit. We define hand-to-mouth status as

$$H = \mathbf{1}\{0 \leq m \leq y/2\} + \mathbf{1}\{m < 0,\ m + \bar{m} \leq y/2\}, \qquad \bar{m} = y. \tag{2}$$

The first branch identifies small nonnegative balances. The second identifies borrowers with little remaining room under the assumed limit, which equals one month of income. Negative net liquid wealth alone does not determine hand-to-mouth status; the borrower branch also depends on the distance from the assumed credit boundary.

Net illiquid wealth separates the two hand-to-mouth groups without changing the liquidity test. With gold outside the balance sheet, poor hand-to-mouth households have $H = 1$ and $a \leq 0$, while wealthy hand-to-mouth households have $H = 1$ and $a > 0$. Non-hand-to-mouth households have $H = 0$. "Wealthy" therefore means positive net illiquid wealth in this classification, rather than membership in a high-income percentile. When gold is counted as illiquid, $a + g$ replaces a at the poor–wealthy boundary.

A net-worth test would combine balance-sheet positions that play different roles in current spending. A household with little liquidity and a home can be wealthy hand-to-mouth even when its net worth is large. The poor–wealthy distinction therefore compares two balance sheets, not two income levels. Both groups are financially constrained under the liquidity rule; what separates them is positive net illiquid wealth.

4.2 Predicting balance sheets in CPHS

We construct CPHS balance sheets by predicting the components that its income and consumption panel lacks. Separate models fitted in AIDIS predict liquid assets, liquid debt, illiquid assets, and illiquid debt from the harmonized variables. We apply these models to CPHS and subtract debts from the corresponding assets. A separate gold model supplies the value used in the placement and collateral-capacity comparisons. Observed income then sets each household's liquidity threshold.

Training and testing data separate model fitting from evaluation. For wealth prediction, AIDIS supplies both datasets because it records the balance-sheet amounts we seek to predict. We estimate each model on the training data and evaluate its predictions for households in the separate testing data. The models use the same household split and survey weights. The training data also determine the ownership cutoff, the correction from log predictions to rupees, and the quantile map; these are then applied to the testing data. Income prediction follows the same distinction within CPHS, where income is observed.

Two-part prediction distinguishes owning an asset from owning a small amount. The first model predicts whether a component is positive. The second predicts its logarithm among positive holders. We correct the conversion from logs to rupees using training residuals. Gradient boosting allows these relationships to vary across household characteristics. The combined prediction equals the predicted amount when the ownership probability exceeds a calibrated cutoff and zero otherwise. We choose the cutoff to match ownership in the training data and keep it fixed in the testing data. Preserving zeros matters because the poor–wealthy distinction depends on whether net illiquid wealth is positive.

Quantile mapping restores the spread of predicted positive amounts. Conditional predictions tend to cluster too closely together. We rank them within the training prediction distribution and assign the corresponding observed AIDIS amount quantiles. This preserves their ordering while changing their spread. The map learned in the training data is then applied to AIDIS testing data and to CPHS.

Participation mapping adjusts how many CPHS households are predicted to hold each component. We map the CPHS ownership-score distribution to the AIDIS reference distribution before applying the calibrated cutoff. The amount map changes the spread of positive holdings; the participation map changes ownership. Both matter for a classification based on zero holdings and small liquid balances.

Prediction performance differs across the two parts of the model and across balance-sheet components. Table 6 compares six methods in the AIDIS testing data and separates ownership accuracy from amount fit for gradient boosting. Ownership accuracy ranges from 0.884 for illiquid assets to 1.000 for illiquid debt, while the log-amount R^2 among positive holders ranges from 0.398 to 0.560. The combined predictions fit debt positions more closely than asset amounts: gradient boosting has an IHS-R^2 of 0.912 for liquid debt and 0.986 for illiquid debt, compared with 0.043 for liquid assets and 0.014 for illiquid assets. The table also reports rank correlation after quantile mapping, distinguishing the ordering of households from the accuracy of predicted amounts.

Table 6: Prediction of balance-sheet aggregates in AIDIS testing data

	Liquid assets	Liquid debt	Illiquid assets	Illiquid debt
Panel A. Combined predictions: testing-data R^2 (IHS)				
Wealth bins	-0.320	-0.999	-0.805	-1.529
Consumption bins	-0.315	-2.624	-0.215	-9.458
Two-part elastic net	-0.002	0.887	0.038	0.979
Two-part random forest	0.155	0.912	0.029	0.984
Two-part gradient boosting (selected)	0.043	0.912	0.014	0.986
Quantile-gradient-boosting draw	-0.164	0.907	0.011	0.980
Panel B. Two-part gradient boosting: ownership, amounts, and ranks				
Ownership accuracy	0.992	0.988	0.884	1.000
Positive-amount R^2 (log)	0.421	0.398	0.429	0.560
Rank correlation, quantile-mapped predictions	0.644	0.945	0.645	0.996

Notes: Survey-weighted statistics evaluated on AIDIS testing data, using models fitted on separate training data. A combined prediction is the conditional positive amount when predicted ownership exceeds its calibrated threshold and zero otherwise. IHS denotes the inverse hyperbolic sine; negative R^2 indicates fit worse than a constant on that scale. Ownership accuracy is the fraction whose predicted positive-holding status matches observed status. Positive-amount R^2 evaluates log amounts among observed positive holders. Rank correlation is the weighted correlation between the ranks of observed amounts and quantile-mapped combined predictions, including zeros. Training data determine the ownership cutoff, the correction from log predictions to rupees, and the amount map. Gradient boosting is the selected conditional model; its amounts and participation margin are mapped for application to CPHS as described in the text.

Prediction must locate the classification boundaries as well as reproduce asset amounts. The liquidity test compares net liquid wealth with an income-based rupee threshold. The poor–wealthy distinction turns on whether net illiquid wealth is positive. We therefore check whether predicted portfolios place households on the same side of these boundaries as their observed portfolios.

Amount fit and classification agreement answer different questions. In the AIDIS testing data, binary agreement with the observed-wealth liquidity label is 66.89 percent for the conditional prediction and 68.68 percent with the amount map. Agreement on the three-way poor, wealthy, and non-hand-to-mouth label is 64.43 and 63.97 percent, respectively. Appendix Table 13 re-

ports these classification diagnostics because locating the liquidity boundary and locating the illiquid-wealth boundary are distinct prediction tasks.

The gold model distinguishes ownership from the amount held. Its ownership agreement in the AIDIS testing data is 0.833, compared with 0.831 for an always-owner benchmark; the deployed CPHS non-owner share is 17.52 percent. Appendix Table 14 also reports dwelling diagnostics. Table 15 compares gold distributions across surveys: overall non-ownership matches, while predicted positive holdings in CPHS have a higher median. Both margins matter for the collateral-capacity classification.

Income alone does not distinguish the poor and wealthy hand-to-mouth groups. Table 7 describes AIDIS households using observed balance sheets and predicted monthly income. Median predicted income differs much less across groups than their portfolios do. Poor and wealthy hand-to-mouth households both have small liquid balances. Median illiquid wealth is Rs. 661,000.00 in the wealthy group and zero in the poor group. Non-hand-to-mouth households instead stand out for their median liquid wealth of Rs. 18,500.00. The table also reports borrowing and residence patterns.

Table 7: Balance sheets and characteristics by financial-constraint status in AIDIS

Characteristic	Poor HtM	Wealthy HtM	Non-HtM
Share of households (%)	7.19	44.58	48.23
Median net liquid wealth (Rs.)	1,500.00	1,160.00	18,500.00
Median net illiquid wealth (Rs.)	0.00	661,000.00	850,000.00
Has outstanding borrowing (%)	26.79	55.10	28.78
Urban household (%)	68.01	24.34	37.23
Median predicted monthly income (Rs.)	11,792.93	12,242.88	12,594.85
Households in characteristic summaries	7,311	52,332	56,813

Notes: HtM denotes financially constrained households under the liquidity rule. AIDIS households under the benchmark combined saver–borrower classification, with gold outside both wealth aggregates and an assumed borrowing limit of one month of income. Wealth is observed; monthly income is predicted using the gradient-boosting model fitted in CPHS. Shares and medians use AIDIS survey weights. Wealth and income are in rupees. Group shares refer to the classification sample of 116,461 households. The final row reports the observations included in each group's weighted characteristic summaries.

The CPHS classification depends on how the wealth model represents amounts and ownership. Table 8 reports the classification under alternative wealth-prediction methods, holding the Kaplan–Violante–Weidner benchmark boundary fixed. All methods use observed CPHS income and the same combined saver–borrower rule. Gold remains outside both wealth aggregates in this comparison.

Quantile mapping and the stochastic draw raise the wealthy hand-to-mouth share relative to conditional gradient-boosting predictions. The wealthy share is 25.82 percent under gradient boosting, 45.03 percent with quantile mapping, and 45.53 percent under the quantile-gradient-

boosting draw. Mapping participation as well as positive amounts gives a wealthy share of 40.74 percent and a poor share of 6.20 percent in the main specification. The comparison shows why both the spread of predicted amounts and the ownership margin matter when the balance sheet is translated into household groups.

Table 8: Direct CPHS classification by wealth-prediction method

Prediction method	Poor HtM	Wealthy HtM	Non-HtM
Gradient boosting	1.60	25.82	72.58
Gradient boosting, quantile-mapped	3.27	45.03	51.70
Quantile map plus participation map (main)	6.20	40.74	53.06
Quantile-gradient-boosting draw	3.10	45.53	51.36

Notes: HtM denotes financially constrained households under the liquidity rule. Entries are weighted percentages of 129,906 CPHS households in the fixed 2019 classification. All rows use observed positive monthly income, gold outside both wealth aggregates, the combined saver–borrower rule, and a borrowing limit of one month of income. Wealth models are fitted on AIDIS training data and applied to CPHS, using ownership cutoffs calibrated in AIDIS. The main specification maps both positive amounts and participation. Rounding can cause rows not to sum to exactly 100.

The asset-placement comparisons can also be repeated on the predicted CPHS balance sheets. Table 9 reports these comparisons under the main prediction method, with gold entering through its separate prediction. The rows retain the same income measure and classification rule while changing which assets and debts enter liquid and illiquid wealth. The real-estate screen changes only the illiquid side. The gold rows place the same asset on either side of the liquidity boundary, while the broader definitions also include other assets and a wider set of debts.

Counting a broader set of assets as liquid raises the non-hand-to-mouth share beyond the gold-only case. The non-hand-to-mouth share is 90.67 percent when the broader bundle enters liquid wealth, compared with 84.41 percent when only gold is added. Counting the broader bundle as illiquid and extending both debt aggregates gives a non-hand-to-mouth share of 50.39 percent. These comparisons distinguish the resources counted as available for current spending from those counted as illiquid wealth. The next subsection examines the gold-placement comparison in detail.

Table 9: Hand-to-mouth shares in CPHS under alternative balance-sheet boundaries

Definition	Poor HtM	Wealthy HtM	Non-HtM
KVW benchmark	6.20	40.74	53.06
Benchmark with real-estate screen	6.30	40.64	53.06
Gold included in illiquid wealth	1.30	45.64	53.06
Gold included in liquid wealth	1.70	13.89	84.41
Broader physical assets liquid	1.42	7.90	90.67
Broad illiquid wealth and debt	1.00	48.61	50.39

Notes: HtM denotes financially constrained households under the liquidity rule. Weighted percentages of 129,906 CPHS households in the fixed 2019 classification. Income is observed; wealth uses gradient boosting with quantile and participation mapping and separate gold prediction. All rows use the combined saver–borrower rule and a borrowing limit of one month of income. The benchmark excludes gold and the broader asset bundle from both wealth aggregates. The real-estate screen removes forest, water and other excluded land, farm structures, unfinished buildings, wells and other constructions from illiquid wealth. It also removes agricultural land and dwelling-plus-residential-land bundles valued below Rs. 100,000. The broader liquid bundle contains gold, transport equipment, livestock, agricultural and non-farm machinery, and post-office saving certificates. The broad illiquid row adds this bundle to illiquid assets. It also extends liquid debt to business operating expenses, debt repayment and loans in kind, and illiquid debt to business capital expenditure and financial investment.

4.3 CPHS prevalence and the placement of gold

Most financially constrained households in CPHS are wealthy hand-to-mouth under the benchmark definition. Table 9 reports shares of 40.74 percent wealthy hand-to-mouth and 6.20 percent poor hand-to-mouth when gold is outside both aggregates. The non-hand-to-mouth share is 53.06 percent. Positive illiquid wealth and a small liquid buffer therefore coexist for a substantial part of the classification sample.

Counting gold as illiquid changes the poor–wealthy division while preserving the liquidity classification. The poor share becomes 1.30 percent and the wealthy share becomes 45.64 percent. The non-hand-to-mouth share remains 53.06 percent because neither net liquid wealth nor the threshold has changed. Gold ownership can thus distinguish households by wealth even when it contributes nothing directly to the measured liquid buffer.

Counting gold as fully liquid changes the liquidity boundary itself. The non-hand-to-mouth share rises to 84.41 percent, while the poor and wealthy hand-to-mouth shares become 1.70 and 13.89 percent. This definition assigns the entire predicted gold value to liquid wealth. The comparison establishes the importance of gold's placement in the accounting rule; it does not describe an observed sale or pledge of gold.

Alternative classifications in AIDIS show the same pattern: counting assets as liquid reduces the share classified as financially constrained. Appendix Table 18 reports the six definitions using observed wealth and income predicted from CPHS. The real-estate screen and the placement of gold in illiquid wealth change the poor–wealthy division while leaving total hand-to-mouth at 51.77 percent. Counting gold as liquid reduces this share to 22.33 percent; including the

broader liquid-asset bundle reduces it to 12.92 percent. Counting the broader bundle as illiquid and extending the debt aggregates instead gives a total hand-to-mouth share of 54.70 percent. These definitions ask which assets can finance current expenditure and which liabilities are counted against them.

4.4 Potential liquidity from gold

Collateral capacity measures whether part of the gold position is sufficient to bridge the liquid shortfall. Starting from the gold-illiquid classification, let λ be the fraction of gold value counted as pledgeable and let $g_{\min}$ be a minimum eligible holding. With $\widetilde{g} = g\mathbf{1}\{g \geq g_{\min}\}$, a collateral-capable wealthy hand-to-mouth household satisfies

$$H = 1, \qquad a + g > 0, \qquad m + \lambda\widetilde{g} > y/2. \tag{3}$$

The main rule sets $\lambda = 0.75$ and $g_{\min} = 0$. These are classification parameters, not observed household credit limits or loan approvals.

Gold ownership alone does not establish sufficient collateral capacity. The rule values an eligible holding at a fraction of its value and asks whether that amount would bridge the household's liquid shortfall. A small holding can fail this condition even when it is eligible to be pledged. Gold remains in illiquid wealth, while its assumed pledgeable value enters the separate capacity test. The test partitions financially constrained households with positive illiquid wealth without changing who passes the original liquidity rule.

The capacity rule divides wealthy hand-to-mouth households into two sizable groups. Table 10 reports 28.18 percent of all CPHS households as wealthy hand-to-mouth with sufficient gold capacity and 17.46 percent as wealthy hand-to-mouth without it. Poor hand-to-mouth households account for 1.30 percent, and non-hand-to-mouth households for 53.06 percent. The capacity distinction preserves the original gold-illiquid liquidity status while describing an additional portfolio margin.

The capacity rule also divides the AIDIS wealthy hand-to-mouth group into two sizable groups. The AIDIS row of Table 10 uses observed wealth and predicted income. It reports 26.69 percent of all households as wealthy hand-to-mouth with sufficient gold capacity and 22.39 percent as wealthy hand-to-mouth without it. The grouped columns show how these two categories divide the total wealthy hand-to-mouth share while leaving poor and non-hand-to-mouth status unchanged.

Table 10: Financial constraints and potential gold collateral capacity

Survey	Poor HtM	Wealthy HtM			Non-HtM
		Total	Without sufficient gold capacity	With sufficient gold capacity	
CPHS	1.30	45.64	17.46	28.18	53.06
AIDIS	2.70	49.08	22.39	26.69	48.23

Notes: HtM denotes financially constrained households under the liquidity rule. Entries are survey-weighted percentages of all households in each survey. The wealthy-HtM total is divided between the two gold-capacity columns. CPHS uses the fixed 2019 classification of 129,906 households, with observed monthly income and predicted wealth. AIDIS covers 116,461 households, with observed wealth and monthly income predicted using the gradient-boosting model fitted in CPHS. Gold is included in illiquid wealth in both surveys; AIDIS also includes paintings and other artistic originals. Both rows use the combined saver–borrower rule and an assumed borrowing limit of one month of income. Sufficient gold capacity requires $m + 0.75g > y/2$, with no minimum gold floor. Capacity is a portfolio classification, not observed borrowing.

The annual analysis compares constrained households with and without sufficient gold capacity. The group without capacity combines poor hand-to-mouth households and wealthy hand-to-mouth households that fail the capacity condition. We compare it with collateral-capable wealthy hand-to-mouth and non-hand-to-mouth households. We then divide the group without capacity by net illiquid wealth excluding gold: $a \leq 0$ versus $a > 0$. This asks whether other illiquid wealth distinguishes responses when gold alone cannot bridge the shortfall.

Alternative valuation and eligibility rules change the measured size of the capable group. With no minimum gold floor, its share is 25.14 percent at a pledgeable fraction of 0.50 and 29.14 percent at a fraction of 0.85. At the main fraction of 0.75, a Rs. 10,000 floor gives a share of 26.53 percent. Appendix B reports the full grid and the complementary AIDIS classifications.

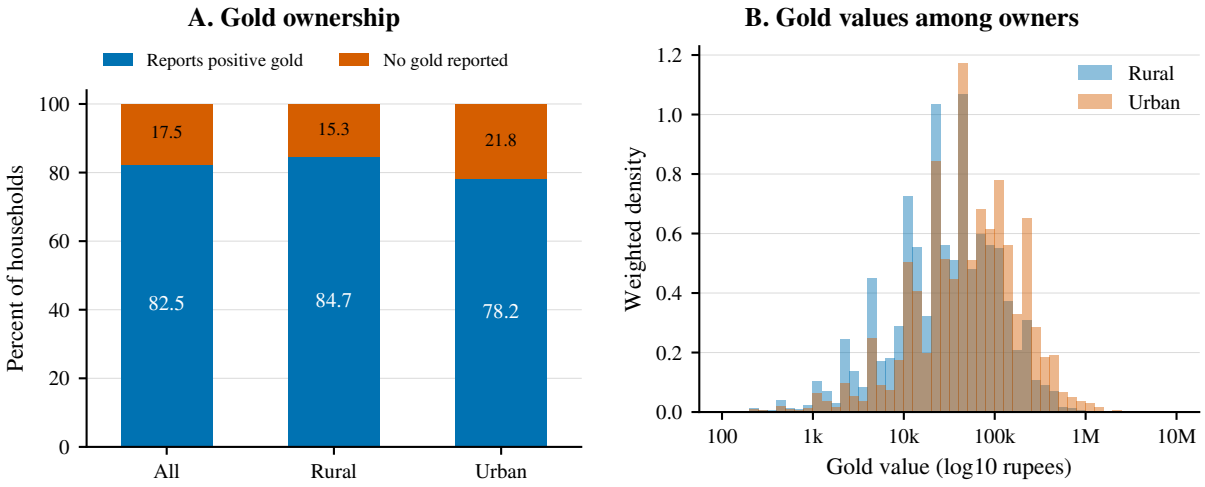
We keep the 2019 household classification fixed when estimating annual responses. A household is not reassigned to a different group as its annual income changes. This lets us compare income-shock responses across the initial portfolio groups.

4.5 Gold in Indian household finance

Gold combines saving with social and consumption uses in India. An ornament can be worn at a wedding, passed down to a child, or used to raise money. As Singh et al. [2026] emphasize, jewellery can be both a cherished possession and a financial asset. In August 2026, the World Gold Council [2026] estimated Indian household gold holdings at about 31,000 tonnes. Households may therefore have substantial wealth in gold while also having reasons to retain particular pieces.

Gold is widely held in both rural and urban India, but the amounts differ. Panel A of Figure 1 shows positive gold holdings for 84.7 percent of rural households and 78.2 percent of urban households in the descriptive AIDIS sample. Panel B shows that, among owners, the urban distribution lies further to the right. Gold ownership therefore does not imply the same potential cash buffer for every household. The amount held matters for whether collateral can cover a liquidity shortfall.

Figure 1: Gold ownership and holdings in rural and urban India



Notes: AIDIS 2019, survey-weighted descriptive gold sample: 116,437 households, including 69,439 rural and 46,998 urban households. Panel A includes households reporting no gold. Panel B shows the distribution of positive gold values on a logarithmic rupee scale. This descriptive sample is separate from the classification and annual-response samples.

Gold can remain attractive when formal financial products are costly or difficult to use. A bank account need not provide a savings product suited to irregular earnings, small contributions, and sudden expenses. The Household Finance Committee [2017] documents travel, paperwork, product complexity, and other participation costs. A familiar local jeweller can offer a more understandable way to buy, exchange, or sell gold [World Gold Council, 2023]. The usefulness of financial services matters alongside account ownership.

Preserving purchasing power is another motive for holding gold. Research on Indian gold demand describes households' desire to protect savings against inflation and their familiarity with gold as reasons for choosing it [World Gold Council, 2023]. This motive does not make the return risk-free: gold's rupee price fluctuates, and Erb and Harvey [2013] find that it is an unreliable inflation hedge over practical investment horizons. India also issues Treasury bills and government bonds, which individuals can access through the RBI's Retail Direct scheme [Reserve Bank of India, 2023]. Gold's role in household saving therefore concerns the appeal and accessibility of different assets, rather than an absence of government securities.

Gold also connects accumulated savings to short-term finance. Selling an ornament converts it into cash. Pledging it for a loan can raise cash while allowing the household to recover it after repayment [Singh et al., 2026]. Both routes can help meet urgent funding needs [World Gold Council, 2023]. Our capacity test captures the potential collateral route: the assumed pledgeable value must be large enough to bridge the measured liquid shortfall.

Gold matters both for measuring wealth and for assessing potential liquidity. Its ownership can move a constrained household from the poor to the wealthy group. Its collateral value can then distinguish households within the wealthy group. Section 5 asks whether these portfolio differences are associated with different consumption responses to permanent and transitory income shocks.

5 Annual income shocks and consumption responses

The annual analysis asks whether households with different portfolios respond differently to income shocks. A temporary income loss can be spread over time using savings or borrowing. A lasting loss changes the resources available over many years. We use the BPP approach of Blundell et al. [2008] to distinguish these shocks from the joint behavior of income and consumption. We estimate their pass-through separately across household groups, keeping the 2019 labels fixed.

5.1 Annual outcomes and the income process

Annual outcomes measure the household's income and usual spending over the year. We require at least six months with both income and expenditure observed. Annual income is twelve times average observed monthly income. Annual consumption is twelve times median observed monthly total expenditure. The median reduces the influence of an unusually large purchase in one month. Missing months are not assigned zero income or expenditure.

We remove predictable income and consumption changes before estimating shock responses. The first-stage regressions include household composition and working-age characteristics, religion and caste interacted with year, and year-by-district-by-urban effects. These controls separate common demographic and local changes from the residual changes used by BPP. Appendix C describes the tail and household-history filters. Changes in residual annual logarithms supply the growth moments, and the reported counts refer to the usable estimation samples.

The income process separates a persistent component from a finite-lived transitory component. Write residual log income as $\widetilde{y}_{it} = p_{it} + v_{it}$, where $p_{it} = p_{i,t-1} + \zeta_{it}$. The permanent innovation ζ_{it} changes the persistent level. The transitory component v_{it} is a finite moving average of innovations ε_{it}. Income growth therefore satisfies

$$\Delta\widetilde{y}_{it} = \zeta_{it} + \Delta v_{it}. \tag{4}$$

This distinction allows a temporary disturbance to affect more than one annual observation while keeping it separate from a lasting income change.

Consumption growth has separate loadings on the permanent and transitory innovations. For residual log consumption $\widetilde{c}_{it}$, the response equation is

$$\Delta\widetilde{c}_{it} = \phi\zeta_{it} + \psi\varepsilon_{it} + \Delta u^{c}_{it}, \qquad u^{c}_{it} = u_{it} + \rho u_{i,t-1}. \tag{5}$$

Here ϕ is the response to the permanent shock, and ψ is the response to the transitory innovation. The additional consumption component allows measurement noise or transitory consumption variation beyond the two income innovations.

The response coefficients measure how strongly consumption adjusts to each income innovation. A log-income innovation of 0.01 changes log consumption by 0.01ϕ if permanent and 0.01ψ if transitory. In this log specification, the coefficients are elasticities. A rupee marginal propensity to consume (MPC) would also depend on consumption relative to income. We report the estimated proportional responses throughout. The transitory innovation's later contribution to income depends on its moving-average terms.

Permanent and transitory shocks leave different patterns in income growth. A temporary increase raises income now and reverses as its effect fades. A permanent innovation has no such finite reversal. BPP uses these time patterns to separate the components. The co-movement of consumption growth with earlier and later income growth then reveals how spending responds to each.

We estimate the model by matching its implied time patterns to the annual data. Diagonally weighted minimum distance uses year-specific variances, autocovariances, and cross-covariances through five annual leads. We include both directions of the cross-covariances. Each group uses 193 moments, and shock variances can vary by year. Household-level sandwich standard errors account for repeated observations from the same household. The resulting coefficients describe consumption adjustment within each group.

5.2 Responses by collateral capacity

Collateral-capable wealthy hand-to-mouth households have the lowest estimated transitory response in the main comparison. Table 11 reports $\psi = 0.1874$ for this group, 0.2192 for constrained households without sufficient capacity, and 0.2063 for non-hand-to-mouth households. Their standard errors are 0.0039, 0.0058, and 0.0027. The constrained groups have little financial liquidity, yet their spending adjusts differently to a temporary income innovation. This is why the potential liquidity of their other assets matters for the comparison.

Table 11: Annual consumption responses by gold collateral capacity

	HtM without capacity	Capable wealthy HtM	Non-HtM
Permanent, ϕ	0.6975***	1.0351***	0.6613***
	(0.0140)	(0.0154)	(0.0059)
Transitory, ψ	0.2192***	0.1874***	0.2063***
	(0.0058)	(0.0039)	(0.0027)
Households	19,938	31,315	75,473
Household-years	137,443	227,504	538,984
Moments	193	193	193

Notes: HtM denotes financially constrained households under the liquidity rule. Annual CPHS, fixed 2019 gold-illiquid classification and age-group demographics. Diagonally weighted minimum distance; 193 moments per group. BPP coefficients are proportional annual consumption responses to income innovations. Parentheses contain household-level sandwich standard errors; significance levels are * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Counts are unweighted estimation counts. HtM without capacity combines poor HtM and all wealthy HtM failing $m + 0.75\widetilde{g} > y/2$. Capable wealthy HtM satisfy that condition.

Permanent-shock pass-through reverses the ordering of the two constrained groups. It is 1.0351 for collateral-capable wealthy households, 0.6975 for constrained households without capacity, and 0.6613 for non-hand-to-mouth households. Their standard errors are 0.0154, 0.0140, and 0.0059. The model estimates these coefficients freely, including values above one. A portfolio associated with a smaller response to a temporary change need not be associated with a smaller response to a lasting change.

The annual estimates use each group's available income and consumption histories. The group without capacity contains 19,938 households, the collateral-capable group 31,315, and the non-hand-to-mouth group 75,473. Table 11 also reports household-years. These are unweighted estimation counts, not the weighted prevalence shares in Section 4. We next ask whether other illiquid wealth matters within the group without gold capacity.

5.3 Responses within the group without capacity

Other illiquid wealth distinguishes responses even among constrained households without gold capacity. Table 12 divides this group by net illiquid wealth excluding gold. Transitory pass-

through is 0.2627 when that wealth is nonpositive and 0.2085 when it is positive. The standard errors are 0.0130 and 0.0064. Pooling the households into one group without capacity would conceal this difference in their estimated responses.

Table 12: Annual responses of financially constrained households without gold capacity

	Net illiquid wealth excluding gold	
	Nonpositive ($a \leq 0$)	Positive ($a > 0$)
Permanent, ϕ	0.7606***	0.6803***
	(0.0285)	(0.0160)
Transitory, ψ	0.2627***	0.2085***
	(0.0130)	(0.0064)
Households	4,465	15,473
Household-years	31,466	105,977
Moments	193	193

Notes: HtM denotes financially constrained households under the liquidity rule. Annual CPHS, fixed 2019 gold-illiquid classification and age-group demographics. Diagonally weighted minimum distance; 193 moments per group. BPP coefficients are proportional annual consumption responses to income innovations. Parentheses contain household-level sandwich standard errors; significance levels are * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Counts are unweighted estimation counts. The first column combines poor HtM and non-capable gold-inclusive wealthy HtM with nonpositive non-gold illiquid wealth; the second contains non-capable wealthy HtM with positive non-gold illiquid wealth. These two columns partition the no-capacity group in Table 11.

Permanent-shock responses also differ within the group without capacity. The estimate is 0.7606 for households with nonpositive non-gold illiquid wealth and 0.6803 for those with positive non-gold illiquid wealth. Their standard errors are 0.0285 and 0.0160, and the groups contain 4,465 and 15,473 households. This split distinguishes limited liquid resources from the absence of other illiquid wealth.

The annual comparisons link consumption adjustment to several features of the household balance sheet. Financial liquidity determines hand-to-mouth status. Gold value determines potential collateral capacity. Other illiquid wealth separates households within the group without that capacity. These are differences associated with portfolios; they do not identify a causal collateral effect or establish that households borrowed against gold. We next examine whether the response patterns persist under alternative balance-sheet definitions.

5.4 Robustness

The robustness exercises vary the balance-sheet definition, household grouping, and annual estimator. Appendix Table 20 compares four definitions: gold excluded, gold illiquid, gold liquid, and a broader illiquid-wealth and debt definition. Each uses fixed 2019 household labels and annual minimum-distance estimation. These poor, wealthy, and non-hand-to-mouth partitions differ from the capacity groups and retain their own samples. For example, the first two def-

initions have 74,718 non-hand-to-mouth households, compared with 75,473 in Table 11. We compare their patterns rather than treating the estimates as coming from one common sample.

The pooled annual estimate provides a reference for the group comparisons. Appendix Table 19 reports transitory pass-through of 0.2011 and permanent pass-through of 0.7657. The transitory estimate is close to the non-hand-to-mouth estimate of 0.2063 in the main capacity comparison. Pooling therefore gives a useful sense of the overall response, while the group estimates show how that response varies with household portfolios.

Poor hand-to-mouth households have the largest transitory-response estimate whether gold is excluded from wealth or counted as illiquid. With gold outside both aggregates, pass-through is 0.2312 for poor HtM, 0.1914 for wealthy HtM, and 0.2053 for non-HtM. When gold is instead illiquid, the poor-group estimate rises to 0.3144, while the wealthy and non-hand-to-mouth estimates are 0.1969 and 0.2053. The ordering is unchanged: poor HtM has the highest point estimate, non-HtM is next, and wealthy HtM has the lowest. Counting gold as wealth changes who belongs to the poor and wealthy groups, but does not make the wealthy group's transitory response exceed the non-hand-to-mouth response.

Counting gold as liquid changes the ordering of wealthy and non-hand-to-mouth responses. The gold-liquid estimates are 0.2669 for poor HtM, 0.2174 for wealthy HtM, and 0.1997 for non-HtM, so both hand-to-mouth groups now have higher transitory pass-through than the non-hand-to-mouth group. This definition allows the full gold position to enter the liquidity test and changes which households remain classified as financially constrained. By contrast, the broad illiquid-wealth and debt definition preserves the gold-illiquid ordering, with estimates of 0.3173, 0.1946, and 0.2071 for poor, wealthy, and non-hand-to-mouth households. The location of gold relative to the liquidity boundary therefore matters more for this ordering than broadening the illiquid balance sheet.

Pooling households by non-gold illiquid wealth preserves the same transitory-response ordering. Appendix Table 21 combines poor HtM with wealthy HtM who lack both gold capacity and positive non-gold illiquid wealth. With gold illiquid, their response is 0.2613, compared with 0.1964 for the remaining wealthy HtM group and 0.2053 for non-HtM. Broadening illiquid wealth and debt gives 0.2592, 0.1943, and 0.2071. The ranking therefore survives both the broader asset definition and a different grouping of financially constrained households.

Households without gold provide a separate check on the role of usable wealth. An annual covariance-ratio specification in Appendix Table 22 uses real per-capita outcomes and excludes the pandemic window. Among predicted gold non-owners, transitory responses are 0.264 for

poor HtM, 0.292 for wealthy HtM, and 0.225 for non-HtM. The pooled HtM–non-HtM difference is 0.064, with a bootstrap standard error of 0.028. Both financially constrained groups have higher point estimates when gold cannot provide a buffer. This comparison uses its own sample and estimator; it complements the minimum-distance results above.

The robustness results support the main finding that constrained households need not have the same transitory response. The gold-illiquid wealthy group combines households with and without capacity. Separating them reveals pass-through of 0.1874 for collateral-capable wealthy households and 0.2192 for the constrained group without capacity. Dividing the latter group by non-gold illiquid wealth reveals a further difference: 0.2627 when that wealth is nonpositive and 0.2085 when it is positive. Across these comparisons, portfolio composition distinguishes households that all have limited financial liquidity.

Permanent-shock rankings are more sensitive to the household classification. With gold excluded, pass-through is 1.0917 for poor HtM and 0.9184 for wealthy HtM. Counting gold as illiquid reverses that ordering: the estimates become 0.7195 and 0.9906. Non-hand-to-mouth pass-through remains 0.6998. The broad illiquid definition also gives wealthy HtM the higher response, while the gold-liquid definition gives poor HtM the higher response. The conclusion draws together these findings about wealth, liquidity, and shock persistence.

6 Conclusion

We construct a joint household income and balance-sheet dataset for India and use it to connect financial constraints to consumption responses. Harmonizing CPHS and AIDIS places predicted assets and debts alongside observed income and consumption in CPHS. This resource supports two exercises: measuring the shares of hand-to-mouth households and estimating their responses to income shocks using BPP.

Gold changes what the balance sheet tells us about financial constraints. Most constrained households in the benchmark classification already own positive illiquid wealth. Counting gold as illiquid moves more households from poor to wealthy hand-to-mouth without changing their liquid resources. Its potential collateral value then separates wealthy constrained households according to whether gold could bridge their liquid shortfall. Ownership, liquidity, and collateral capacity describe different aspects of the same portfolio.

Collateral capacity distinguishes consumption responses within the financially constrained population. Wealthy hand-to-mouth households with sufficient gold capacity have lower estimated

transitory-shock pass-through than constrained households without it. Within the latter group, households without positive non-gold illiquid wealth have the higher response. Permanent shocks produce a different ordering. The composition of wealth therefore matters alongside the persistence of the income change when describing household consumption adjustment.

The robustness results reinforce the importance of distinguishing household portfolios. Poor hand-to-mouth households have the highest transitory-response point estimate under all four annual definitions. The ordering of wealthy and non-hand-to-mouth households changes when gold counts as liquid, and permanent-response rankings also vary. The separate annual covariance-ratio check also distinguishes capacity groups and finds higher hand-to-mouth responses among predicted gold non-owners. These comparisons show what is lost when financial constraints are summarized by a single poor–wealthy distinction.

Combining existing surveys offers other countries a practical way to study household financial constraints. The first step is to bring income and balance-sheet information together using common measures. Classification can then reflect the assets households actually own, and panel data can connect those portfolios to consumption adjustment. Future research could use asset sales and borrowing transactions to study how households turn potential liquidity into resources for spending.

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A Prediction and validation

The additional prediction checks examine classification boundaries, separate asset models, and income prediction. Section 4.2 and Table 6 report the wealth-model comparison in AIDIS testing data. The tables below evaluate how the predictions support classification and the complementary income exercise.

Threshold diagnostics evaluate the predicted portfolio where it determines group membership. Table 13 compares observed and predicted poor, wealthy, and non-hand-to-mouth shares in the AIDIS testing data. Binary agreement refers only to whether a household is financially constrained under the liquidity rule; three-way agreement also requires the correct sign of net illiquid wealth. The quantile and participation route has the same testing-data entries as the quantile route because the participation map is an identity on the reference distribution.

Table 13: Classification diagnostics in the AIDIS testing data

Route	Poor	Wealthy	Non-HtM	Binary	Three-way
Observed wealth	7.15	44.02	48.83	—	—
Conditional prediction	2.84	24.63	72.53	66.89	64.43
Quantile map	7.64	51.48	40.89	68.68	63.97
Quantile + participation	7.64	51.48	40.89	68.68	63.97
Quantile draw	6.75	45.88	47.37	60.11	56.36

Notes: All entries are percentages. Poor and wealthy refer to HtM groups. Binary agreement compares the liquidity label with the label obtained using observed wealth; three-way agreement compares the full poor/wealthy/non-HtM label, using the same income input. The participation map is an identity on the AIDIS reference distribution.

Asset-specific diagnostics distinguish gold from the dwelling check. Table 14 reports ownership agreement in the AIDIS testing data and the always-owner benchmark for each target, together with the predicted CPHS non-owner share. Gold is predicted separately for classification and capacity. The dwelling check evaluates an asset already included in illiquid wealth, so its predicted value is not added to that aggregate again.

Table 14: Gold and dwelling ownership diagnostics

Target	Testing-data agreement	Always-owner	CPHS non-owner (%)
Gold	0.833	0.831	17.52
Dwelling	0.871	0.816	18.17

Notes: Agreement and the always-owner benchmark are fractions in the AIDIS testing data; the deployed CPHS non-owner share is a percentage. The dwelling is already part of illiquid wealth.

The gold prediction preserves overall ownership while allowing rural and urban shares to differ. Table 15 reports non-ownership of 17.52 percent in both surveys. Predicted CPHS non-

ownership is higher than observed AIDIS non-ownership in rural areas, 16.05 versus 15.33 percent, but lower in urban areas, 20.50 versus 21.83 percent. Among owners, the median is Rs. 50,000 in CPHS and Rs. 38,500 in AIDIS; the corresponding means are Rs. 77,854 and Rs. 74,272. Thus, matching the overall ownership margin does not force the same positive holdings in both surveys. The ownership decision determines who has gold, while the amount held determines its potential collateral value.

Table 15: Observed and predicted gold distributions

Measure	AIDIS observed	CPHS predicted
No gold, all households (%)	17.52	17.52
No gold, rural households (%)	15.33	16.05
No gold, urban households (%)	21.83	20.50
Median gold among owners (Rs.)	38,500	50,000
Mean gold among owners (Rs.)	74,272	77,854

Notes: Survey-weighted distribution summaries. AIDIS records gold values; CPHS values come from the separate ownership and positive-amount models. Non-ownership shares use all households in the relevant survey or rural–urban group. Means and medians condition on positive gold. These compare observed AIDIS and predicted CPHS distributions; ownership accuracy in AIDIS testing data is reported separately in Table 14.

Income prediction supplies the liquidity threshold for the complementary AIDIS classification. We fit the income models in CPHS, where total income is observed, and evaluate them in separate CPHS testing data. Table 16 compares the age-profile benchmark, elastic net, random forest, and gradient boosting. Baseline gradient boosting has a testing-data R^2 of 0.810 and a Spearman rank correlation of 0.891. These diagnostics concern predicted log income; the main CPHS exercise instead requires wealth prediction. Appendix B examines the resulting household classifications.

Table 16: Total-income prediction in the CPHS testing data

Model	RMSE	MAE	R^2	Rank corr.
Age-profile benchmark	0.386	0.301	0.581	0.753
Elastic net	0.303	0.229	0.741	0.854
Random forest	0.281	0.209	0.777	0.871
Gradient boosting	0.260	0.195	0.810	0.891
Reweighted boosting	0.272	0.205	0.792	0.880
Boosting without consumption	0.305	0.229	0.737	0.851

Notes: Testing data: 25,981 households. RMSE and MAE refer to log income; rank correlation is Spearman correlation. The target is total income, and the reported statistics use CPHS testing data. These models supply predicted income for the AIDIS classification.

B Classification checks

The capacity grid asks how gold valuation and eligibility change potential liquidity among financially constrained households. Table 17 reports the share of all households that are wealthy hand-to-mouth and collateral-capable under each rule. It uses gold as illiquid wealth and the same CPHS prediction method and liquidity test as the main text. Each cell changes only the pledgeable fraction and the minimum gold holding shown in its row and column.

Table 17: Sensitivity of CPHS collateral-capable shares

Minimum gold (Rs.)	$\lambda = 0.50$	0.65	0.75	0.85
0	25.14	27.07	28.18	29.14
5,000	24.92	26.77	27.84	28.77
10,000	23.91	25.58	26.53	27.37

Notes: Weighted percentage of all 129,906 classification households who are wealthy HtM and collateral-capable. The gold-illiquid definition, two-part wealth prediction with quantile and participation mapping, and KVW liquidity rule remain fixed. Columns vary the pledgeable fraction; rows vary the minimum gold holding.

AIDIS provides a complementary classification with observed wealth and predicted income. Table 18 compares six definitions. The benchmark gives 7.19 percent poor HtM and 44.58 percent wealthy HtM. Counting gold as illiquid changes these shares to 2.70 and 49.08 percent, while non-hand-to-mouth remains 48.23 percent. Counting gold as liquid raises the non-hand-to-mouth share to 77.67 percent. These are separate prevalence results from CPHS because the survey and prediction direction differ. Table 10 reports the corresponding capacity partition. Appendix C turns to the annual response comparisons.

Table 18: Hand-to-mouth shares in AIDIS under alternative balance-sheet boundaries

Definition	Poor HtM	Wealthy HtM	Non-HtM	Total HtM
KVW benchmark	7.19	44.58	48.23	51.77
Benchmark with real-estate screen	9.47	42.31	48.23	51.77
Gold included in illiquid wealth	2.70	49.08	48.23	51.77
Gold included in liquid wealth	3.55	18.78	77.67	22.33
Broader physical assets liquid	2.73	10.19	87.08	12.92
Broad illiquid wealth and debt	1.99	52.70	45.30	54.70

Notes: HtM denotes financially constrained households under the liquidity rule. Weighted percentages of 116,461 AIDIS households. Wealth is observed and monthly income is predicted using the gradient-boosting model fitted in CPHS. All rows use the combined saver–borrower rule and a borrowing limit of one month of income. Total HtM includes poor and wealthy hand-to-mouth households. The asset and debt definitions and the real-estate screen follow Table 9. The AIDIS gold-illiquid and broad illiquid aggregates also include paintings and other artistic originals. Rounding can cause displayed shares not to add exactly to the reported total.

C Annual robustness checks

C.1 Pooled annual specification

The pooled annual specification provides a reference for the group response estimates. Table 19 reports $\phi = 0.7657$ and $\psi = 0.2011$, with household-level sandwich standard errors of 0.0059 and 0.0018. Consumption adjusts more strongly to a lasting income innovation than to a temporary one. The table also reports the income-persistence and consumption-noise parameters, using 193 moments.

Table 19: Pooled annual consumption responses

Parameter	BPP response
Permanent, ϕ	0.7657***
	(0.0059)
Transitory, ψ	0.2011***
	(0.0018)
First transitory lag, θ_1	0.2512***
	(0.0031)
Second transitory lag, θ_2	0.0492***
	(0.0026)
Consumption-noise persistence, ρ	0.1084***
	(0.0028)
Consumption-noise variance, σ_u^2	0.0195***
	(0.0001)
Moments	193

Notes: BPP-response column only. Annual growth moments for 2015–2025; diagonally weighted minimum distance. Income has an MA(2) transitory component and consumption noise is MA(1) in levels. Shock variances vary by year. Parentheses contain household-level sandwich standard errors. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. This pooled specification is distinct from the capacity-group estimates in Tables 11–12.

The transitory process permits an innovation to affect income beyond its initial year. It uses $v_{it} = \varepsilon_{it} + \theta_1 \varepsilon_{i,t-1} + \theta_2 \varepsilon_{i,t-2}$, with estimates 0.2512 and 0.0492 for the two lag coefficients. The consumption-noise component is moving-average of order one in levels. These process parameters belong to this pooled specification and are not substituted into the capacity-group estimates of Section 5.

The annual sample uses continuing household spells with usable income and consumption histories. Annual levels require at least six paired months. We trim the extreme one percent tails of annual log outcomes within year, remove histories with income growth above 500 percent or below minus 80 percent, and exclude late-entry Sikkim, Meghalaya, and Tripura. Each covariance uses households observed at its required dates.

C.2 Alternative balance-sheet definitions

The group estimates ask whether responses depend on the balance-sheet definition. Table 20 reports four annual poor/wealthy/non-hand-to-mouth partitions: gold outside wealth, gold illiquid, gold liquid, and broad illiquid wealth and debt. Each retains its own sample. Section 5.4 compares their findings with the capacity results.

Table 20: Annual consumption responses under alternative balance-sheet definitions

	Poor HtM	Wealthy HtM	Non-HtM
Panel A: gold outside both aggregates			
Permanent, ϕ	1.0917***	0.9184***	0.6998***
	(0.0273)	(0.0155)	(0.0073)
Transitory, ψ	0.2312***	0.1914***	0.2053***
	(0.0063)	(0.0034)	(0.0026)
Households	12,607	38,309	74,718
Household-years	93,633	269,200	534,521
Moments	193	193	193
Panel B: gold illiquid			
Permanent, ϕ	0.7195***	0.9906***	0.6998***
	(0.0388)	(0.0144)	(0.0073)
Transitory, ψ	0.3144***	0.1969***	0.2053***
	(0.0186)	(0.0030)	(0.0026)
Households	2,537	48,379	74,718
Household-years	16,923	345,910	534,521
Moments	193	193	193
Panel C: gold liquid			
Permanent, ϕ	0.7811***	0.6837***	0.7708***
	(0.0349)	(0.0203)	(0.0068)
Transitory, ψ	0.2669***	0.2174***	0.1997***
	(0.0142)	(0.0071)	(0.0021)
Households	3,586	12,763	109,285
Household-years	24,961	86,497	785,896
Moments	193	193	193
Panel D: broad illiquid wealth and debt			
Permanent, ϕ	0.7512***	0.9414***	0.7030***
	(0.0530)	(0.0130)	(0.0075)
Transitory, ψ	0.3173***	0.1946***	0.2071***
	(0.0224)	(0.0029)	(0.0027)
Households	1,810	52,237	71,587
Household-years	11,935	373,189	512,230
Moments	193	193	193

Notes: HtM denotes financially constrained households under the liquidity rule. Annual CPHS, fixed 2019 classifications and age-group demographics. The balance-sheet definitions follow Table 9. Each group is estimated separately by diagonally weighted minimum distance, with an MA(2) transitory income component, MA(1) consumption noise in levels and year-specific shock variances. Coefficients are proportional annual consumption responses to income innovations. Parentheses contain household-level sandwich standard errors. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Counts are unweighted and refer to each panel's own estimation sample. These poor/wealthy/non-HtM partitions differ from the capacity groups in Tables 11–12.

C.3 Alternative portfolio groupings

The portfolio-pooling exercise asks whether the results depend on separating poor HtM from households whose illiquid wealth consists only of gold. Table 21 pools these groups when gold cannot bridge their liquid shortfall. The comparison group combines collateral-capable wealthy HtM with other wealthy HtM who have positive non-gold illiquid wealth. Both panels use the annual minimum-distance estimator. The first counts gold as illiquid; the second broadens illiquid assets and debts.

The first pooled group has the highest transitory response under both definitions. Its estimates are 0.2613 and 0.2592, compared with 0.1964 and 0.1943 for the second group. Permanent responses reverse that ranking: the second group's estimates are 0.9914 and 0.9410, compared with 0.7972 and 0.8482 for the first. The distinction between a temporary shortfall and a lasting loss of resources therefore remains important under this alternative grouping.

Table 21: Annual responses under alternative portfolio groupings

	Poor HtM + non-capable, $a \leq 0$	Capable wealthy HtM + non-capable, $a > 0$	Non-HtM
Panel A. Gold included in illiquid wealth			
Permanent, ϕ	0.7972***	0.9914***	0.6998***
	(0.0334)	(0.0148)	(0.0073)
Transitory, ψ	0.2613***	0.1964***	0.2053***
	(0.0125)	(0.0031)	(0.0026)
Households	4,428	46,488	74,718
Household-years	31,266	331,567	534,521
Moments	193	193	193
Panel B. Broad illiquid wealth and debt			
Permanent, ϕ	0.8482***	0.9410***	0.7030***
	(0.0435)	(0.0132)	(0.0075)
Transitory, ψ	0.2592***	0.1943***	0.2071***
	(0.0150)	(0.0030)	(0.0027)
Households	3,053	50,994	71,587
Household-years	21,349	363,775	512,230
Moments	193	193	193

Notes: Annual CPHS minimum-distance estimates using fixed 2019 household groups and growth years 2015–2025. Non-capable denotes wealthy hand-to-mouth households without sufficient gold collateral capacity; a is net illiquid wealth excluding gold. The first column pools poor HtM with non-capable wealthy HtM whose $a \leq 0$. The second pools collateral-capable wealthy HtM with non-capable wealthy HtM whose $a > 0$. Capacity uses a pledgeable fraction of 0.75 and no gold floor. The income process has a second-order moving-average transitory component and year-specific shock variances. Household-level sandwich standard errors are in parentheses. These samples belong to the alternative balance-sheet specifications in Table 20, separately from the main capacity comparison. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.4 Annual covariance-ratio checks

A covariance-ratio estimator provides a separate annual check on transitory responses. It divides the covariance of consumption growth with next-period income growth by the income-growth autocovariance. Temporary shocks reverse, allowing the ratio to isolate their consumption response. This exercise uses twelve-month windows of residual real per-capita log outcomes, fixed 2019 classifications, and excludes March 2020–June 2021.

The full-sample annual estimates again separate wealthy households by gold capacity. Panel A of Table 22 gives responses of 0.101 with capacity and 0.167 without. Their difference is 0.067, with a standard error of 0.019. This preserves the main capacity ordering under a different estimator and sample.

The gold-free subsample asks whether financially constrained households respond more when gold cannot provide a buffer. Both hand-to-mouth groups in Panel B have higher point estimates than non-HtM. The pooled-HtM difference is 0.064, with a standard error of 0.028. Standard errors use 500 household-cluster bootstrap replications and account for covariance between group estimates.

Table 22: Annual covariance-ratio checks

Household group or contrast	Annual response (s.e.)
Panel A. Full classification sample	
Poor HtM	0.114 (0.022)
Collateral-capable wealthy HtM	0.101 (0.011)
Wealthy HtM without capacity	0.167 (0.016)
Non-HtM	0.140 (0.005)
All households	0.135 (0.005)
Without capacity − with capacity	+0.067 (0.019)
Wealthy HtM without capacity − non-HtM	+0.027 (0.017)
Collateral-capable wealthy HtM − non-HtM	−0.040 (0.012)
Poor and wealthy HtM without capacity − non-HtM	+0.012 (0.014)
Panel B. Predicted gold non-owners	
Poor HtM	0.264 (0.063)
Wealthy HtM	0.292 (0.027)
Non-HtM	0.225 (0.014)
All households	0.242 (0.012)
All HtM − non-HtM	+0.064 (0.028)
Wealthy HtM − non-HtM	+0.067 (0.030)

Notes: Annual covariance-ratio estimates using consecutive twelve-month windows of residual real per-capita log outcomes and fixed 2019 participation-mapped classifications. The sample excludes March 2020–June 2021. Gold capacity uses a pledgeable fraction of 0.75 and no gold floor. Panel B restricts the sample to predicted gold non-owners. Standard errors use 500 stratified household-cluster bootstrap replications; contrast standard errors account for covariance between group estimates. These proportional responses use a separate sample and estimator from the main minimum-distance results.